SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: **Total Marks: 10**

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

```
A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>
```

Snippet 2: Form Submission Methods

```
A. GET
B. POST
C. PUT
D. DELETE
```

Snippet 3: Navigation Elements

```
A. <div>
B. <nav>
C. <section>
D. <header>
```

Snippet 4: XML Syntax

```
A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note
```

Snippet 5: Web Communication Protocols

```
A. FTP
B. HTTP
C. SMTP
D. SNMP
```

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

Code of Conduct:

I Vaishnavee Bhavani Vijayakumar (192571071) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

TECHNICAL CASE STUDY & SOLUTIONS

Web Development Standards Implementation

QUESTIONS, DATA SNIPPETS & COMPREHENSIVE SOLUTIONS

Scenario: A start-up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols. The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippet	Provided Options
Snippet 1: Video Embedding	**A.** <media src="video.mp4"></media> | **B.** <video src="video.mp4" controls></video> **C.** <movie src="video.mp4"></movie> | **D.** <vid src="video.mp4"></vid>
Snippet 2: Form Submission	**A.** GET | **B.** POST | **C.** PUT | **D.** DELETE
Snippet 3: Navigation Elements	**A.** | **B.** <nav> | **C.** <section> | **D.** <header>
Snippet 4: XML Syntax	**A.** <note><to>User</to><from>Admin</from> **B.** <note><to>User</to><from>Admin</from></note> **C.** <note><to>User</from></note> | **D.** <note to="User"></note
Snippet 5: Web Protocols	**A.** FTP | **B.** HTTP | **C.** SMTP | **D.** SNMP

Q1 **Select the correct HTML tag for video embedding and justify your answer.**

Correct Choice: **Option B:** `<video src="video.mp4" controls></video>`

Justification: The `<video>` element is the standard native HTML5 specification for embedding video media directly into web documents without requiring third-party plugins. The boolean attribute `controls` enables built-in playback interfaces (play, pause, volume, and seek bars). Elements like `<media>`, `<movie>`, and `<vid>` are non-standard and unrecognized by web browsers.

Q2 **Identify the appropriate HTTP method for form submission and explain why.**

Correct Choice: **Option B: POST**

Explanation: The registration form collects user credentials and personal details. The `POST` method packages submitted data within the body of the HTTP request rather than appending it to the URL query string (as `GET` does). This prevents sensitive data from being visible in the address bar, saved in browser history, cached by proxies, or exposed in server access logs. It also has no URL-imposed length constraints.

Q3 **Choose the proper semantic element for navigation and justify its usage.**

Correct Choice: **Option B:** `<nav>`

Justification: The `<nav>` element represents a landmark section of a page intended specifically for navigation links (menus, tables of contents, or primary link lists). Unlike generic non-semantic containers like `<div>`, `<nav>` communicates clear structural intent to assistive technologies (screen readers) for direct skipping and informs search engine crawlers of key site pathways. While `<header>` often encloses navigation, `<nav>` is the definitive element for navigation links.

Q4 **Identify the correct XML syntax and explain the error in other options.**

Correct Choice: **Option B:** `<note><to>User</to><from>Admin</from></note>`

Justification: Option B is well-formed XML: it has an enclosed root element (`<note>`), properly nested child tags, and valid case-sensitive opening and closing delimiters.

Analysis of Errors in Other Options:

- **Option A (`<note><to>User</to><from>Admin</from>`):** Missing closing root tag. XML strictly requires all opened tags, including root elements, to have matching closing tags (`</note>`).
- **Option C (`<note><to>User</from></note>`):** Mismatched tag pairing. The opening tag `<to>` is incorrectly terminated by a closing tag `</from>`, violating the nesting rule.
- **Option D (`<note to="User"></note`):** Malformed closing delimiter. The closing tag is incomplete because it is missing the mandatory terminating right angle bracket (`>`).

Q5 **Choose the correct protocol for web communication and justify your choice.**

Correct Choice: **Option B: HTTP (Hypertext Transfer Protocol) / HTTPS**

Justification: HTTP (and its encrypted variant HTTPS) is the foundational request-response application-layer protocol of the World Wide Web. It governs the transmission of hypertext documents, media assets, stylesheets, scripts, and API payloads between client web browsers and web servers. In contrast:

- **FTP (File Transfer Protocol):** Designed strictly for bulk file uploads/downloads between host filesystems.
- **SMTP (Simple Mail Transfer Protocol):** Used exclusively for transmitting email messages between mail servers.
- **SNMP (Simple Network Management Protocol):** Used for monitoring and configuring network devices (routers, switches).

Implementation Summary: Valid modern web delivery requires standard HTML5 media containers (`<video controls>`), secure payload transmission via `POST` over `HTTP(S)` , accessible semantic hierarchy with `<nav>` , and strictly well-formed root/child pairs for XML data interchanges.